

CarbonLens

Find where a factory's carbon leaks.
Fix it for less than it costs.

Industrial Emission Leak-Point Detector
& Circular Alternative Recommender

TEAM NEBULA NUKE

Kunal Ramanuja · Team Lead

Zeel Nakum · Backend

Yash Chauhan · Frontend

Akshat Govindiya · AI/ML Engineer

3 screens

input → hotspots → actions

2 numbers

₹ saved + tCO₂e cut per fix

0 black boxes

cited factors, explainable rules

Small factories can't see their own carbon.

Buyers, banks and regulators now ask MSMEs for a carbon number. A typical 100-person factory has no ESG team, no measured data, and no budget for consultants. Emissions stay invisible, and circular options – alternative materials, recycling loops, process changes – are never evaluated because nobody can put a cost on them.

6.3 Cr

MSMEs in India; >90% have never measured emissions

Scope 3

large buyers must now report supplier emissions under BRSR

₹0

typical SME budget for sustainability consulting

Who it's for

SMEs & factory operators

Get a footprint number and a 3-action plan in 10 minutes – plain language, ₹ lakh units, works on a phone.

Sustainability consultants

Serve 10× more clients; export a methodology-backed report per site.

Industry regulators & large buyers

Onboard a supplier base for Scope 3 with GHG-Protocol-aligned, auditable numbers.

What the brief asks for – and what we deliver

Requirement in problem statement	How CarbonLens covers it
Business inputs process data: energy source, materials, waste streams	Guided 3-step form with unit-aware fields + industry presets (textile, foundry, food)
System identifies top emission sources	Emission-factor engine ranks every source; Pareto chart; 'Leak Point #1' callout; Scope 1/2/3 split
Recommends specific circular interventions – alternative materials, recycling loops, process changes	Rule library tagged Substitute · Loop · Optimise, fired per source above a share threshold
...with estimated cost and CO₂ savings	Every card: capex, ₹ saved/yr, tCO ₂ e cut/yr, payback months, confidence tag
Users: SMEs, factory operators, consultants, regulators	Plain-language UI · multi-client export · cited methodology page

Three steps. Two numbers. One plan.

1 Input

What you burn, buy and throw away – 3 short screens. Presets for textile, foundry and food so the demo never starts blank.

2 Detect

Every source ranked by tCO₂e. One chart makes the leak point obvious – usually a single boiler or grid power.

3 Fix

Costed circular interventions sorted by payback. Add to plan → before/after footprint, ₹ total, exportable PDF report.

CarbonLens Input Hotspots Actions

Step 1 of 3 · Energy

Preset: Textile · Foundry · Food

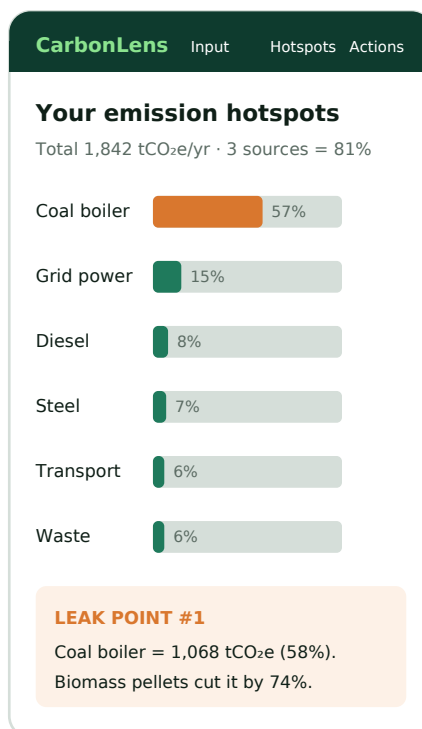
Grid electricity

Diesel generator

Coal-fired boiler

[+ Add another source](#)

Next: Materials →



CarbonLens Input Hotspots Actions

Recommended fixes

Biomass pellet boiler

Substitute

₹18.5 L capex **–790 tCO₂e/yr**

₹9.2 L saved/yr **Payback 24 mo**

Rooftop solar 150 kWp

Substitute

₹60 L capex **–165 tCO₂e/yr**

₹11 L saved/yr **Payback 65 mo**

Closed-loop water reuse

Loop

₹3 L capex **–92 tCO₂e/yr**

₹2.1 L saved/yr **Payback 17 mo**

CarbonLens Input Hotspots Actions Report

Screen 4 · Your circular plan · Ambica Textiles Pvt Ltd

–1,047 t

CO₂e cut / yr

₹22.3 L

cost saved / yr

57%

footprint cut

Before

After

Compliance readiness

- GHG Protocol Scope 1 & 2 aligned
- BRSR-ready summary for buyers
- Every factor cited (CEA · IPCC · DEFRA)
- Methodology page + audit trail

Export PDF report

90-second demo flow

0:00

Pick 'Textile' preset – form pre-fills

0:20

Hotspots animate: coal boiler = 58%

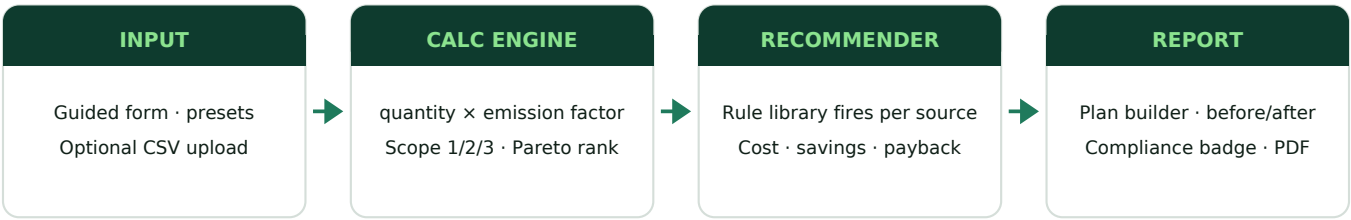
0:45

Add 3 fixes to plan; see ₹ and tCO₂e

1:15

Export report; end on –57% / ₹22 L

Rules for the numbers, AI for the words.



Emission factors (cited: CEA · IPCC · DEFRA)

Source	Unit	kgCO ₂ e/unit	Scope	Circular alternative
Grid electricity (India)	kWh	0.716	2	Rooftop solar, VFDs, LED
Coal (boiler)	tonne	2,420	1	Biomass pellets, heat recovery
Diesel (genset)	litre	2.68	1	Battery storage
Steel (virgin)	tonne	1,850	3	Recycled steel (~600)
Plastic (virgin)	tonne	2,500	3	rPET / returnable crates
Cement	tonne	900	3	Fly-ash / GGBS blend
Road freight	t-km	0.105	3	Load consolidation, rail
Organic waste → landfill	tonne	580	3	Biogas / compost (credit)

Intervention rule (one of 12 in the starter library)

```
{
  "targets": "coal_boiler",
  "fires_if_share": ">= 15%",
  "type": "Substitute",
  "title": "Replace coal boiler with biomass pellet boiler",
  "co2_reduction_pct": 0.74,
  "capex_inr_per_tonne_coal_yr": 7700,
  "opex_delta_inr_per_tonne_coal": -3800,
  "confidence": "High",
  "source": "MNRE Biomass Programme 2024"
}
```

Where AI fits - and where it doesn't

- Rules compute every ₹ and tCO₂e figure, so numbers are auditable and explainable to a regulator.
- The LLM writes the 60-word 'Why this works' and proposes 2 extra process-specific ideas.
- Responses are pre-generated for the presets, so the demo never waits on an API.

Stack: Next.js · Tailwind + shadcn/ui · Recharts (D3) · JSON factor & rule libraries served via REST route · LLM API · Vercel. Deliberately no blockchain, IoT or database – the value is in the recommendation, not the plumbing.

Demo factory: Ambica Textiles, Surat

–57%

footprint after 3 actions (1,842
→ 795 tCO₂e/yr)

₹22.3 L

operating cost saved every
year

24 mo

payback on the biggest fix
(biomass boiler)

Impact goals from the brief

Impact goal	How CarbonLens delivers
Makes emission sources visible and actionable for smaller businesses	Pareto chart + Leak Point callout within 30 seconds of input; no ESG expertise needed
Drives adoption of circular practices through concrete, costed recommendations	Payback-sorted intervention cards; plan roll-up shows total ₹ saved and tCO ₂ e cut
Supports compliance with emerging carbon regulations	GHG-Protocol-aligned method, BRSR-ready summary, every factor cited, exportable PDF

Roadmap

● Hackathon (24 h)

3 industries · 13 factors · 12 rules · PDF export · live on Vercel

● Month 1-3

Import electricity bills & Tally ledgers (no typing) · 20 industries · consultant multi-client view

● Month 4-12

Smart-meter APIs · buyer portal to onboard a supplier base for Scope 3 · push verified reductions to carbon-credit registries

Business model

- Free for SMEs – the footprint number and first plan cost nothing.
- Paid for consultants – multi-client workspace and branded reports.
- Paid for corporate buyers – onboard their whole supplier base for Scope 3 (the wedge).

Team Nebula Nuke

Kunal Ramanuja
Team Lead

Product, pitch, demo flow

Zeel Nakum
Backend

Calc engine, rule library,
REST routes

Yash Chauhan
Frontend

Next.js screens, charts,
PDF export

Akshat Govindiya
AI/ML Engineer

Factor data,
recommender, LLM layer

Circular economy scales when the fix is cheaper than the problem.

CarbonLens proves it is – one factory at a time.